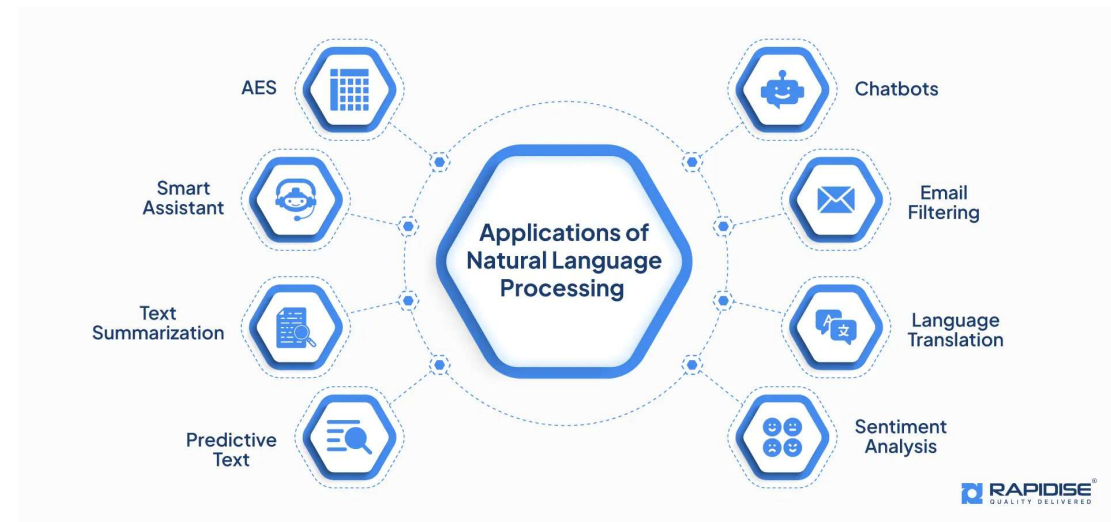


Introduction to NLP

- Varun Raste
Data Scientist

Introduction to NLP

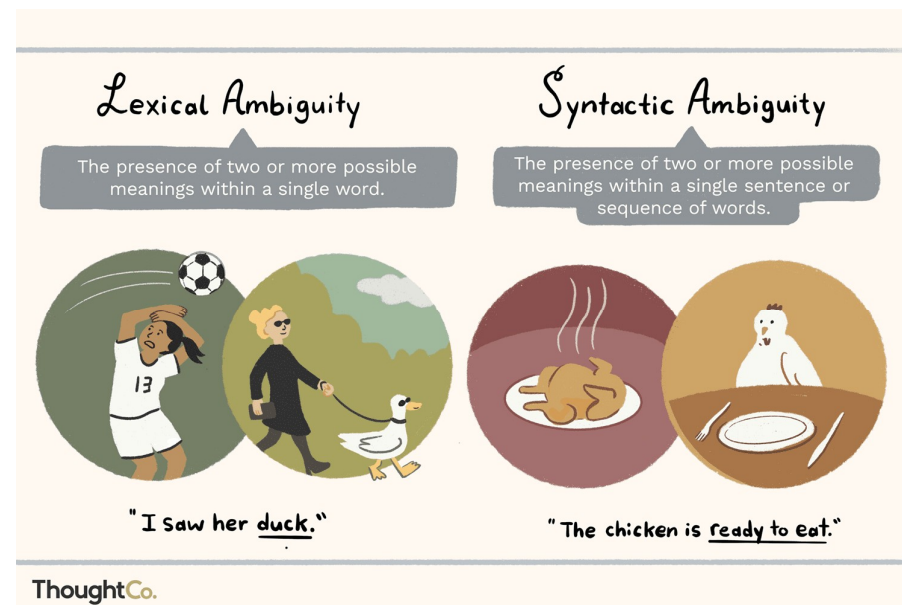
- **Natural Language Processing (NLP)** focuses on enabling machines to read, understand, and generate human language.
- It deals with both written text and spoken language, making it central to modern AI systems.
- NLP converts unstructured language data into structured representations.
- Common applications include chatbots, search engines, recommendation systems, and document analysis.
- The ultimate goal is to extract *meaning*, not just words.



All the images in PPT are taken from internet , author doesn't claim rights*

Why NLP Is Challenging

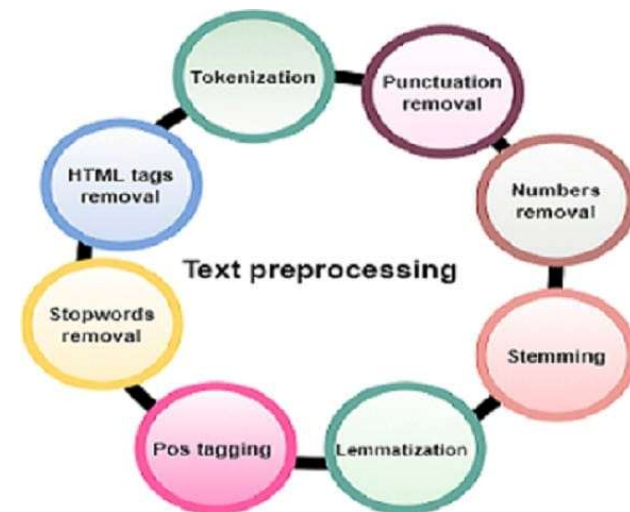
- Human language is inherently ambiguous and context-driven.
- The same word can mean different things depending on usage and sentence structure.
- Grammar rules often break in real-world text such as social media or chats.
- Sarcasm, slang, abbreviations, and spelling errors make NLP harder.
- Machines must learn patterns without explicit linguistic rules.



All the images in PPT are taken from internet , author doesn't claim rights*

Text Processing

- Real-world text data is noisy, inconsistent, and unstructured.
- Preprocessing removes unnecessary elements like punctuation and extra spaces.
- Text normalization ensures consistency across documents.
- Stopword removal helps reduce low-information words.
- Preprocessing improves model efficiency and accuracy.



All the images in PPT are taken from internet , author doesn't claim rights*

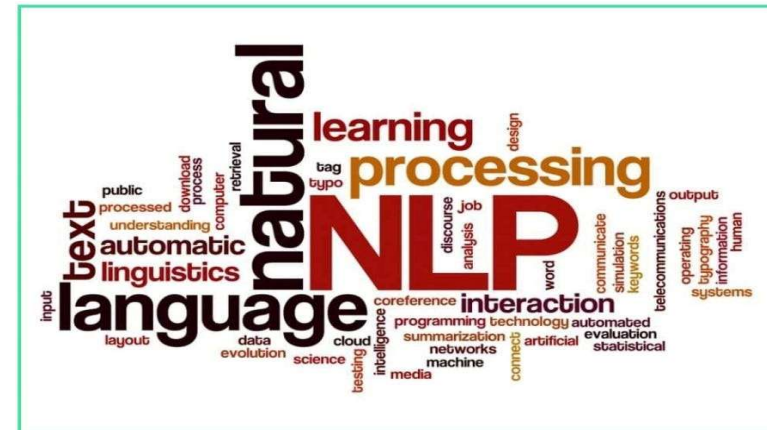
Text Processing – Normalization , Stemming , lemmatization

- Different forms of the same word increase feature space unnecessarily.
- Normalization helps group similar words under a common representation.
- This step directly impacts search quality and model performance.

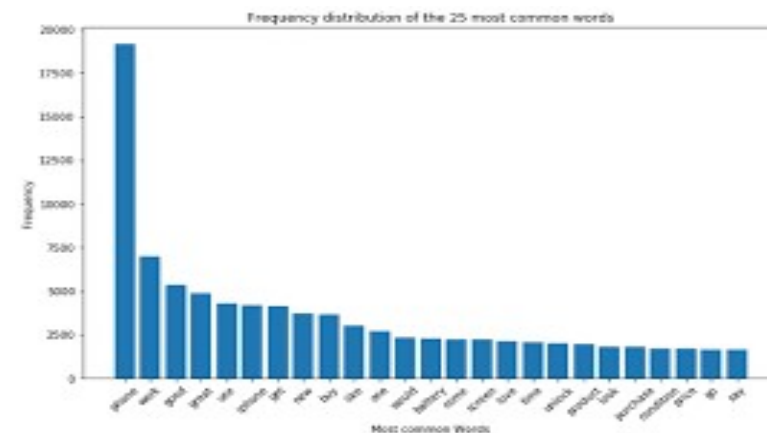
Aspect	Stemming	Lemmatization
Core Idea	Cuts word suffixes mechanically	Uses linguistic rules & vocabulary
Output	May produce invalid words	Produces valid dictionary words
Speed	Very fast	Slower than stemming
Accuracy	Lower	Higher
Example	<i>running</i> → <i>runn</i>	<i>running</i> → <i>run</i>
Usage	Search, indexing	Semantic analysis, NLP models

Text Visualization & EDA – Wordcloud & Top-K words

- WordCloud visualizes word frequency intuitively.
- Larger words indicate higher occurrence.
- Useful for quick thematic understanding.
- Best suited for exploration, not metrics.

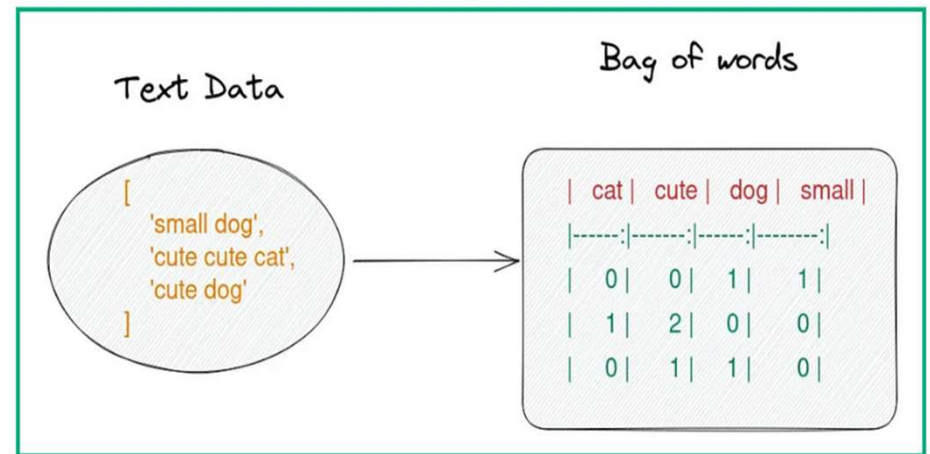


- Lists most common words numerically.
- Helps identify stopwords and domain terms.
- Enables frequency-based filtering decisions.
- More precise than WordCloud.



Vectorization - Bag of Words

- Bag of Words converts text into numeric count vectors.
- Each unique word becomes a feature column.
- Word order and context are ignored.
- Simple and interpretable representation.
- Suffers from high dimensionality and sparsity.



All the images in PPT are taken from internet , author doesn't claim rights*

Vectorization - Term Frequency Inverse Document Frequency

- TF measures how often a word appears in a document.
- IDF reduces the weight of common words across corpus.
- Combines local importance with global rarity.
- Produces more informative features than BoW.
- Widely used in information retrieval systems.

Term Frequency X **Inverse Document Frequency**

$$w_{x,y} = tf_{x,y} \times \log\left(\frac{N}{df_x}\right)$$

Text1: Basic Linux Commands for Data Science

Text2: Essential DVC Commands for Data Science

	basic	commands	data	dvc	essential	for	linux	science
Text 1	0.5	0.35	0.35	0.0	0.0	0.35	0.5	0.35
Text 2	0.0	0.35	0.35	0.5	0.5	0.35	0.0	0.35

All the images in PPT are taken from internet , author doesn't claim rights*

Cosine Similarity

- Measures similarity based on angle between vectors.
- Focuses on direction rather than magnitude.
- Useful when vector lengths vary.
- Common metric for text similarity tasks.
- Works well with TF-IDF and embeddings.

Cosine Similarity in NLP

What's the angle?

How it works: A simple analogy

- High similarity: $\cos(\theta) = 1$
- No similarity: $\cos(\theta) = 0$
- Opposite meaning: $\cos(\theta) = -1$

Example in action

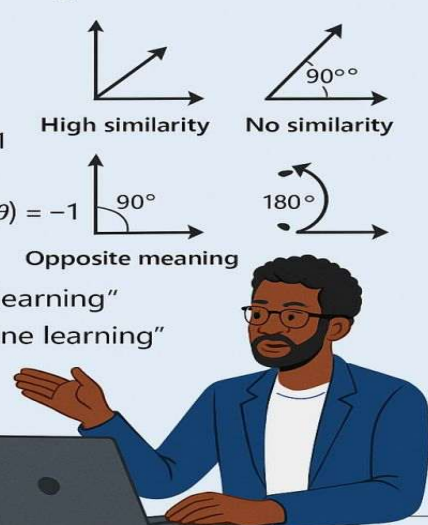
Document 1: "I love deep learning"

Document 2: "I love machine learning"

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

Example aaction

Document 1:
"I love deep learning"



All the images in PPT are taken from internet , author doesn't claim rights*

Search Methods –Keyword Search

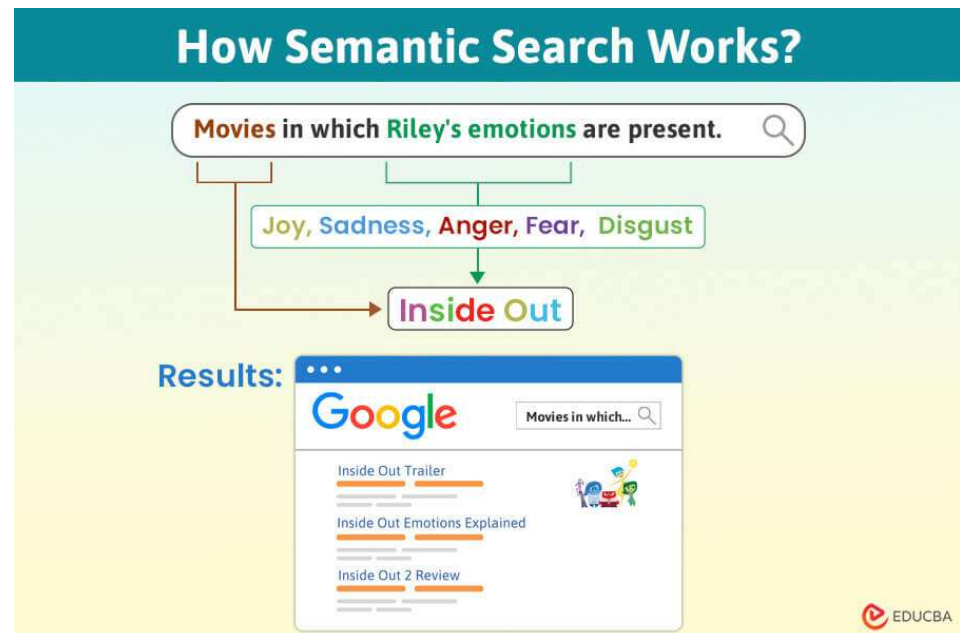
- Retrieves documents based on exact word matches.
- Easy to understand and debug.
- Performs well for structured queries.
- Fails when synonyms or paraphrases are used.
- Traditional search engines rely heavily on it.

Keyword Match Types

Search Term: iPhone Case			
	Syntax	Appears for..	Would show for..
Broad Match	iPhone Case	Searches related to your keyword	Apple phone case, best iPhone case
Phrase Match	"iPhone Case"	Searches with the same meaning as your keyword	Blue iPhone case, iPhone case waterproof
Exact Match	[iPhone Case]	Searches for the same exact meaning or intent as your keyword	iPhone case, iPhone cases, case iPhone

Search Methods –Semantic Search

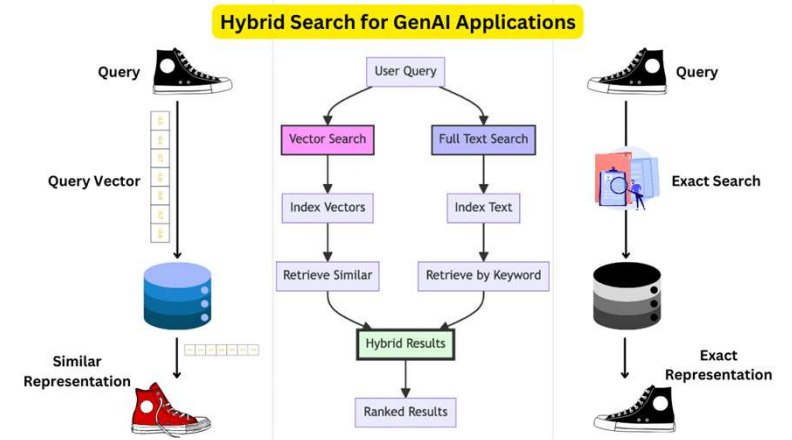
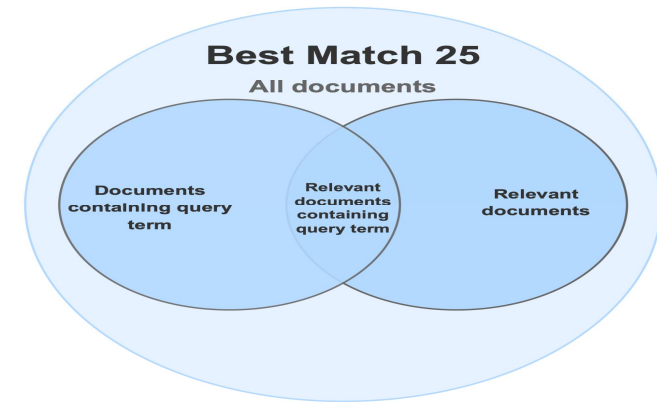
- Understands intent rather than exact wording.
- Uses embeddings to capture semantic meaning.
- Handles paraphrases and synonyms naturally.
- Retrieves conceptually relevant results.
- Used in modern AI-powered search systems.



All the images in PPT are taken from internet , author doesn't claim rights*

BM25 vs Hybrid Search

Aspect	BM25 Search	Hybrid Search
Search Type	Lexical (keyword-based)	Keyword + Semantic
Core Idea	Rank documents using term frequency and rarity	Combine exact match with meaning
Handles Synonyms	✗ No	✓ Yes
Handles Paraphrases	✗ Poorly	✓ Effectively
Uses Embeddings	✗ No	✓ Yes
Interpretability	High (easy to debug)	Medium (fusion logic involved)
Performance	Fast and efficient	Slightly slower but more accurate
Typical Use Case	Traditional search engines, logs	Modern AI search, RAG systems



All the images in PPT are taken from internet , author doesn't claim rights*